Highways of the Generalised Langton's Ants $\mathtt{L}^{K-1}\mathtt{R}$ from the Empty Grid: a Machine-Checked Proof, and a Census up to Twelve Colours

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Abstract

A generalised Langton's ant is given by a rule word of K letters L and R. Gajardo, Lutfalla and Rao proved that the ants $\mathtt{L}^{2k}\mathtt{R}$ ($k > 1$) have highways of periods $16k + 2$ and $32k + 4$, and stated without full proof that the ants $\mathtt{L}^{2k+1}\mathtt{R}$ have a highway of period $32k + 20$. We prove, for every $K \geq 4$, that the ant $\mathtt{L}^{K-1}\mathtt{R}$ started on the empty grid builds a highway from step $24K - 17$ onward, and from no earlier step, with shortest period exactly $16K - 12$ steps, moving two squares diagonally each period. For $K = 2k + 2$ this gives a complete proof of the $\mathtt{L}^{2k+1}\mathtt{R}$ highway of period $32k + 20$, and shows that the empty grid reaches it; for odd $K \geq 5$ it shows that the empty grid reaches the period-$(32k + 4)$ highway of $\mathtt{L}^{2k}\mathtt{R}$. The proof runs the ant once with K kept as a symbol: each square's colour is an affine expression $a + bK$, and each spin of the ant round a 2×2 loop is taken in one step. The part of the run the proof needs is 35 lines long and is printed in full, so it can be checked by hand; a longer certificate, which also gives the exact step at which the ant leaves a fixed box, is checked by an independent program. Separately, we report an experimental census of all 4,015 distinct rules with 2 to 12 colours, sorted by the shape they grow from the empty grid, and some patterns in it. All code and data are public.

Keywords: Langton's ant, generalised ants, turmites, highway, symbolic simulation, computer-assisted proof.
MSC 2020: 37B15, 68Q80.

1 Introduction

Langton's ant [1] moves on an infinite square grid. On a white square it turns right, on a black square it turns left; it then flips the colour of the square and steps forward. From the all-white grid it wanders for about 10,000 steps and then builds a *highway*, a pattern that repeats every 104 steps and moves diagonally away forever. The ant always escapes to infinity [5], but whether it always ends on a highway from every finite starting pattern is open, and the ant can perform universal computation [6], so its long-run behaviour is hard in general.

Turk and Propp generalised the ant to K colours [2, 3, 4]: a rule word $w = w_0 \cdots w_{K-1}$ over $\{\mathtt{L}, \mathtt{R}\}$ says which way to turn on each colour, and the colour then advances by one, modulo K. Gale, Propp, Sutherland and Troubetzkoy [3] showed that rules made of repeated pairs of letters give symmetric patterns.

Recent work. Gajardo, Lutfalla and Rao [9] studied highways of generalised ants by large experiments from random finite starts, and proved highway results by hand. For the family $\mathtt{L}^n\mathtt{R}$ they proved that $\mathtt{L}^{2k}\mathtt{R}$ ($k > 1$) has a *fundamental* highway of period $16k + 2$ and a *harmonic* highway of period $32k+4$, both diagonal; for $\mathtt{L}^{2k+1}\mathtt{R}$ they wrote that "it can be proved analogously"

that there is a highway of period $32k + 20$, and gave a starting pattern, without a full proof. Their proofs decompose the ant's path into *elementary cycles*, which are the spins of this paper. They also showed that some ants, such as LLRLRL, have infinitely many highways. Lutfalla [11] reported that LLLR has, besides its period-52 highway, a rare second highway of period 156, found experimentally; and [10] showed that some ants have both highways and other unbounded behaviours. Gajardo [7] studied the ant through symbolic dynamics (subshifts); our use of "symbolic" is different: we mean simulation with a symbolic parameter.

Contributions.

1. Theorem 3: for every $K \geq 4$, the exact start, shortest period and shift of the highway that $\mathtt{L}^{K-1}\mathtt{R}$ builds from the empty grid. This completes the case $\mathtt{L}^{2k+1}\mathtt{R}$ left as a sketch in [9], and identifies which of the highways of [9] the empty grid reaches.
2. A proof method: one run of the ant with K as a symbol, with spins taken in one step. The part the theorem needs is 35 lines (Table 1); a full certificate of 12,735 lines is verified by an independent checker, in the spirit of the certified proof of the fifth Busy Beaver value [12].
3. An experimental census of all 4,015 distinct rules with $K \leq 12$ colours from the empty grid (Section 5). We did not find such a table in the literature; earlier surveys of simple two-dimensional machines include [8].

The theorem and lemmas are proved; the census describes simulations and depends on our classification tests.

2 Definitions

Definition 1 (Ant)**.** A *rule* is a word $w = w_0 \cdots w_{K-1} \in \{\mathtt{L}, \mathtt{R}\}^K$, $K \geq 2$. A *state* is a position $p \in \mathbb{Z}^2$, a direction $d \in \{0, 1, 2, 3\}$ (up, right, down, left) and a colouring $\sigma : \mathbb{Z}^2 \to \{0, \ldots, K-1\}$ that is 0 on all but finitely many squares. One *step* reads $c = \sigma(p)$, turns right ($d \leftarrow d+1 \bmod 4$) if $w_c = \mathtt{R}$ and left ($d \leftarrow d-1 \bmod 4$) if $w_c = \mathtt{L}$, sets $\sigma(p) \leftarrow c+1 \bmod K$, and moves p one square in direction d. We use screen coordinates: up is $(0, -1)$, right is $(1, 0)$. Unless stated otherwise the ant starts at $p = (0, 0)$, facing up, on the *empty grid* $\sigma \equiv 0$. Steps are numbered from 0.

Every visit adds one to the colour, so $\sigma(p)$ is the number of visits to p so far, modulo K, and on its n-th visit to a square the ant reads $w_{(n-1) \bmod K}$.

Definition 2 (Trace and highway)**.** The *trace* is the sequence $\tau(0), \tau(1), \cdots \in \{\mathtt{L}, \mathtt{R}\}$ of turns made at each step. It is *periodic from step t_0 with period P* if $\tau(t + P) = \tau(t)$ for all $t \geq t_0$. Following [9], the ant is on a *highway* from step t_0 if its trace is periodic from t_0 with some period P and its position after step $t + P$ is its position after step t shifted by a fixed nonzero vector, for all $t \geq t_0$.

We write $A_K = \mathtt{L}^{K-1}\mathtt{R}$. On a square, A_K turns left on each visit except every K-th, where it turns right. So A_K turns right exactly when it reads colour $K - 1$.

3 The theorem

Theorem 3. *For every $K \geq 4$, the ant $A_K = \mathtt{L}^{K-1}\mathtt{R}$ started on the empty grid is on a highway from step $24K - 17$ onward, and its trace is not periodic from any earlier step. The shortest period of its trace is $16K - 12$ steps, and each period moves the ant by $(-2, 2)$. Each period contains 10 right turns, and the numbers of steps between consecutive right turns are, in order,*

$$1,\ 2,\ 4K-3,\ 5,\ 1,\ 1,\ 2,\ 4K-3,\ 4K-7,\ 4K-11.$$

Corollary 4. *For every $k \geq 1$, the ant $\mathtt{L}^{2k+1}\mathtt{R}$ has a highway of period $32k + 20$ whose shift is two squares diagonally, and the empty grid reaches it at step $48k + 31$.*

Proof. Put $K = 2k + 2 \geq 4$ in Theorem 3: $16K - 12 = 32k + 20$ and $24K - 17 = 48k + 31$. □

This is the highway stated in [9, Remark 3], where its proof is only sketched. For $k = 1$ it is the period-52 highway of `LLLR` in [11], which there is observed to start from the empty grid after about 100 steps; Theorem 3 gives step 79.

Corollary 5. *For every $k \geq 2$, the ant $\mathtt{L}^{2k}\mathtt{R}$ started on the empty grid reaches, at step $48k + 7$, a highway of shortest period $32k + 4$ moving two squares diagonally per period.*

Proof. Put $K = 2k + 1 \geq 5$: $16K - 12 = 32k + 4$ and $24K - 17 = 48k + 7$. □

So from the empty grid these ants reach the *harmonic* highway of [9, Theorem 1], not the fundamental one of period $16k + 2$, which [9] reaches from other starting patterns. For $K = 3$ (`LLR`) the theorem does not apply: here the shift by 5 right turns in the list of gaps also fits, because $5 = 4K - 7$ and $1 = 4K - 11$ when $K = 3$. A direct run shows the trace of `LLR` is periodic from step 39 with period 18, moving $(-1, 1)$, which is the fundamental highway of [9].

Direct step-by-step runs, made by the program `proof/highway_concrete.py`, agree with Theorem 3 for $K = 4, 5, 6, 7, 8, 10, 15$ and 20.

Proposition 6 (Leaving a box)**.** *Let $B = \{(x, y) : -1499 \leq x \leq 1498,\ -1499 \leq y \leq 1498\}$. For every $K \geq 4$ the ant A_K first leaves B with the move of step $11{,}980K - 8{,}971$, after exactly 7,488 right turns, made on the same squares in the same directions for every K. The same holds for $K = 3$ (step 26,969).*

Proposition 6 is a consequence of the highway (the ant starts it at step $24K - 17$ and then walks to the edge of B), but we prove it directly from the full certificate, and it was how we first found the family (Section 5.1). The case $K = 3$ is a single ordinary run.

4 The proof

4.1 Symbolic colours

Fix $K \geq 4$. Every number the proof needs is an *affine expression* $a + bK$ with integers a, b. The colour of each visited square is stored as $a + 0K$ with $0 \leq a \leq 2$, or $a + 1K$ with $-3 \leq a \leq -1$; each such expression is a genuine colour in $\{0, \ldots, K - 1\}$ for every $K \geq 4$, and these are the only forms that occur. The ant turns right exactly when the colour is $K - 1$, that is, when the stored expression is $-1 + 1K$: an expression $a + 0K$ with $a \leq 2$ never equals $K - 1$ when $K \geq 4$. All other decisions are comparisons, and every comparison used has the form

$$a + bK \geq c \ \text{ for every } K \geq 4, \quad \text{which holds exactly when } b \geq 0 \text{ and } a + 4b \geq c.$$

4.2 Spins

Lemma 7 (Spin)**.** *Let the ant be at square c_0 facing d, and let c_0, c_1, c_2, c_3 be the squares it visits if it turns left four times (its fourth move returns it to c_0 facing d). Let $r_i = (K - 1) - \sigma(c_i)$, the number of further visits on which c_i is read as* `L`*, and suppose $n = \min_i r_i \geq 1$. Then the next $4n$ steps are n laps of left turns round $c_0, \ldots, c_3$; afterwards the ant is again at c_0 facing d, and each $\sigma(c_i)$ has grown by n.*

Proof. In lap $\ell \leq n$ the square c_i is visited with colour $\sigma(c_i) + \ell - 1 \leq \sigma(c_i) + r_i - 1 = K - 2$, so the ant turns left there. Four left turns bring it back to c_0 facing d, and after n laps each colour is $\sigma(c_i) + n \leq K - 1$. □

The run takes a spin in one step whenever all four r_i are positive for every $K \geq 4$; then n and the $4n$ steps are affine expressions. Otherwise it makes one ordinary step. Positions and directions stay ordinary integers, since no decision depends on the value of K.

4.3 The first 21 right turns

Table 1 lists every action of the run up to the 21st right turn. Each line gives the step at which it starts, the square, the direction the ant faces on arriving, and either the colour read and the turn made, or the number of laps of a spin. The reader can check each line from the ones before it, using Lemma 7 and the rule that colour $K-1$ means turn right. From the table, right turn 11 is at step $24K-17$ on square $(2,1)$, and right turn 21 is at step $40K-29$ on square $(0,3)$; the gaps between right turns $11, 12, \ldots, 21$ are those listed in Theorem 3, and they add up to $16K-12$.

Table 1: Every action of A_K up to its 21st right turn, valid for every $K \geq 4$. A spin of n laps takes $4n$ steps; every other line is one step. "Colour" is the colour read; the number after "right" counts right turns.

Line	Step	Square	Facing	Action	Colour	Turn or laps
1	0	$(0,0)$	up	spin		$K-1$ laps
2	$4K-4$	$(0,0)$	up	step	$K-1$	right (1)
3	$4K-3$	$(1,0)$	right	spin		$K-1$ laps
4	$8K-7$	$(1,0)$	right	step	$K-1$	right (2)
5	$8K-6$	$(1,1)$	down	spin		$K-1$ laps
6	$12K-10$	$(1,1)$	down	step	$K-1$	right (3)
7	$12K-9$	$(0,1)$	left	step	$K-1$	right (4)
8	$12K-8$	$(0,0)$	up	step	$K-1$	right (5)
9	$12K-7$	$(1,0)$	right	step	$K-1$	right (6)
10	$12K-6$	$(1,1)$	down	step	0	left
11	$12K-5$	$(2,1)$	right	step	$K-1$	right (7)
12	$12K-4$	$(2,2)$	down	spin		$K-1$ laps
13	$16K-8$	$(2,2)$	down	step	$K-1$	right (8)
14	$16K-7$	$(1,2)$	left	spin		$K-1$ laps
15	$20K-11$	$(1,2)$	left	step	$K-1$	right (9)
16	$20K-10$	$(1,1)$	up	spin		$K-2$ laps
17	$24K-18$	$(1,1)$	up	step	$K-1$	right (10)
18	$24K-17$	$(2,1)$	right	step	$K-1$	right (11)
19	$24K-16$	$(2,2)$	down	step	$K-1$	right (12)
20	$24K-15$	$(1,2)$	left	step	$K-2$	left
21	$24K-14$	$(1,3)$	down	step	$K-1$	right (13)
22	$24K-13$	$(0,3)$	left	spin		$K-1$ laps
23	$28K-17$	$(0,3)$	left	step	$K-1$	right (14)
24	$28K-16$	$(0,2)$	up	spin		1 lap
25	$28K-12$	$(0,2)$	up	step	$K-1$	right (15)
26	$28K-11$	$(1,2)$	right	step	$K-1$	right (16)
27	$28K-10$	$(1,3)$	down	step	$K-1$	right (17)
28	$28K-9$	$(0,3)$	left	step	1	left
29	$28K-8$	$(0,4)$	down	step	$K-1$	right (18)
30	$28K-7$	$(-1,4)$	left	spin		$K-1$ laps
31	$32K-11$	$(-1,4)$	left	step	$K-1$	right (19)
32	$32K-10$	$(-1,3)$	up	spin		$K-2$ laps
33	$36K-18$	$(-1,3)$	up	step	$K-1$	right (20)
34	$36K-17$	$(0,3)$	right	spin		$K-3$ laps
35	$40K-29$	$(0,3)$	right	step	$K-1$	right (21)

4.4 Periodicity

For the t-th right turn, let p_t be its square, d_t the direction after turning, S_t the colouring just after p_t is updated, and H_t the set of squares visited so far. The motion after a right turn depends only on p_t, d_t and S_t. Fix i and $j = i + P$; let V be the set of squares visited after right turn i up to and including right turn j, let $D = p_j - p_i$, let $f(c) = c \cdot D$, and let $A = (V + D) \setminus V$.

Lemma 8 (Periodicity)**.** *Suppose*

(a) $d_j = d_i$*;*

(b) $S_j(c + D) = S_i(c)$ *for every* $c \in V$*;*

(c) no square of A *is in* H_j*; and* $f(D) > 0$ *and* $\min_{c \in A} f(c) + f(D) > \max_{c \in H_j} f(c)$.

Then for every $k \geq 1$ *the motion from right turn* $i + kP$ *to right turn* $i + (k+1)P$ *is the motion from right turn* i *to right turn* j *moved by* kD*. In particular the trace is periodic from the step of right turn* i*, with period the number of steps from right turn* i *to right turn* j*, and the ant moves away along* D*.*

Proof. We prove by induction on $k \geq 1$:

$$\mathcal{P}(k): \quad p_{i+kP} = p_i + kD, \qquad d_{i+kP} = d_i, \qquad S_{i+kP}(c + kD) = S_i(c) \text{ for all } c \in V,$$
$$H_{i+kP} = H_j \cup \textstyle\bigcup_{l=1}^{k-1} (V + lD).$$

$\mathcal{P}(1)$ is the definition of D together with (a) and (b).

Assume $\mathcal{P}(k)$. Between right turns i and j the ant reads and writes only squares of V, starting from p_i, d_i and the colours S_i on V. By $\mathcal{P}(k)$, at right turn $i + kP$ the ant is at $p_i + kD$, faces d_i, and the colours on $V + kD$ are those of S_i on V moved by kD. So the next P right turns are the motion from i to j moved by kD: the ant visits exactly $V + kD$, ends at $p_i + (k+1)D$ facing $d_j = d_i$, and $S_{i+(k+1)P}(c + kD) = S_j(c)$ for $c \in V$. This gives the position, direction and visited set in $\mathcal{P}(k+1)$. For the colours, take $c \in V$ and put $z = c + (k+1)D$.

If $c + D \in V$, then $S_{i+(k+1)P}(z) = S_j(c + D) = S_i(c)$ by (b).

If $c + D \notin V$, then $c + D \in A$, and z has never been visited. Indeed $f(z) = f(c + D) + kf(D) \geq \min_A f + f(D) > \max_{H_j} f$, so $z \notin H_j$; and if $z \in V + lD$ with $1 \leq l \leq k$, then $c + mD \in V$ with $m = k + 1 - l$. For $m = 1$ this contradicts $c + D \notin V$; for $m \geq 2$, $f(c + mD) \geq \min_A f + f(D) > \max_{H_j} f \geq \max_V f$, a contradiction. So $S_{i+(k+1)P}(z) = 0$. Also $S_i(c) = S_j(c + D) = 0$, by (b) and because $c + D \in A$ is not in H_j. Hence $S_{i+(k+1)P}(z) = S_i(c)$, proving $\mathcal{P}(k+1)$. Finally $f(p_{i+kP}) = f(p_i) + kf(D) \to \infty$. □

4.5 Proof of Theorem 3

Highway. Take $i = 11$ and $P = 10$. From Table 1, $p_{11} = (2, 1)$ and $p_{21} = (0, 3)$, so $D = (-2, 2)$; the ant faces the same way after both, so (a) holds. The set V has 14 squares, and the program `proof/highway.py` checks (b) on them, comparing affine expressions (equal expressions are equal for every K), and (c), which involves only positions. By Lemma 8 the ant is on a highway from step $24K - 17$, the step of right turn 11, with period $(40K - 29) - (24K - 17) = 16K - 12$ steps and shift $(-2, 2)$.

Shortest period. Suppose the trace is periodic, from some step on, with a period $p < 16K - 12$. A word with periods p and q from some point on also has period $\gcd(p, q)$ there, so we may take p to divide $16K - 12$. Shifting by p maps right turns to right turns, so it moves the cyclic list of 10 gaps of Theorem 3 by some s places with $0 < s < 10$, and leaves it unchanged; then it is also unchanged by a shift of $\gcd(s, 10) \in \{1, 2, 5\}$ places. A shift by 1 or 2 would need $1 = 4K - 3$, so $K = 1$. A shift by 5 would need $5 = 4K - 7$, so $K = 3$. For $K \geq 4$ none is possible.

Exact start. The step before right turn 11 is right turn 10 (line 17 of the table: the gap is 1), so $\tau(24K - 18) = \texttt{R}$. One period later, step $24K - 18 + 16K - 12 = 40K - 30$ is the step before right turn 21, and the gap before right turn 21 is $4K - 11 \neq 1$ for $K \geq 4$, so $\tau(40K - 30) = \texttt{L}$. Hence the trace is not periodic from step $24K - 18$, nor from any earlier step. □

4.6 A certificate checked by an independent program

Proposition 6 needs the run up to the edge of B, which is 12,735 actions (8,990 single steps and 3,744 spins), too long to print. The proof program writes them all to `proof/certificate.txt`, in the format of Table 1. A separate program, `proof/check_certificate.py`, which shares no code with it, replays the ant from the empty grid line by line. It makes no choices; it checks that each line's square, direction and colour agree with its own replay; that every colour stays in $\{0, \ldots, K-1\}$; that a step turns `R` exactly when the colour is $K-1$; that every spin satisfies Lemma 7; and that spins near the edge of B are not jumped. This is 68,153 inequalities, each checked for every $K \geq 4$ by the rule above. From its own replay the checker then finds the edge at step $11{,}980K - 8{,}971$ after 7,488 right turns, and re-checks every claim of Theorem 3: conditions (a)–(c) at right turns 11 and 21, the shortest period and the exact start. As a test, we altered the certificate in several ways (a spin one lap longer, a left turn changed to a right turn, a line deleted, a change inside the first 35 lines); each was rejected at the line changed.

5 A census of multi-colour ants

This section is experimental.

Machines that are the same. Swapping every `L` and `R` gives the mirror image of the same path, so we keep one rule of each pair, the one starting with `L`. If $w = u^j$ for a shorter word u (for example `RLRL`), the ant walks the path of u, so we keep only words that are not powers of shorter ones. This leaves $1, 3, 6, 15, 27, 63, 120, 252, 495$, 1,023 and 2,010 machines for $K = 2, \ldots, 12$, 4,015 in all (half the number of primitive binary words of length K).

Method. Every rule word with $2 \leq K \leq 12$ was run for 10^6 steps from the empty grid on a 3000×3000 grid, stopping if the ant left the box B. Each final picture was given one class by these tests, applied in order to the bounding box of the visited squares; the thresholds were tuned by looking at random samples of every class.

Class	Test on the picture after 10^6 steps
Highway	The ant left B, and under 5% of the box it crossed is coloured.
Triangle	Width/height ratio over 0.8; one half of the box, cut along a diagonal, under 3% coloured and the opposite half over 30%.
Symmetric	Colours agree with the picture's 180° turn or a mirror image on over 90% of coloured squares, and under 95% of the box coloured.
Square	Width/height ratio over 0.9, and over 86% of the box coloured (or over 80% with mostly coloured borders).
Chaotic	Everything else.

Results. Table 2 and Figures 1 and 2. About two machines in three grow a chaotic blob; from $K = 7$ on, about one in five grows a square, roughly twice as many as build a highway. Only three triangles appear: `LLRLRRLLLLL` ($K = 11$), and `LLLLLLRRLRLL` and `LLRRRLRRRLLL` ($K = 12$), the last being the mirror image of the known triangle rule `RRLLLRLLLRRR`.

Observation 9 (Even runs and symmetry)**.** Read a rule in a circle and split it into runs of equal letters. All 83 symmetric machines have every run of even length (for example `LLRR`, `LRRL`, `LLLRRL`). Of the 104 machines with this property the other 21 were classed square (16, themselves highly symmetric in colour) or chaotic (5). No symmetric machine occurs for odd K.

This agrees with [3], where such rules are shown to give patterns that are symmetric whenever the ant is back at its start; the 5 "chaotic" ones are most likely pictures taken between two such moments.

Table 2: Distinct machines by number of colours K and class, from the empty grid, after 10^6 steps.

K	Highway	Square	Triangle	Symmetric	Chaotic	Machines
2	1	0	0	0	0	1
3	1	0	0	0	2	3
4	2	0	0	2	2	6
5	1	2	0	0	12	15
6	2	3	0	5	17	27
7	9	12	0	0	42	63
8	13	22	0	10	75	120
9	24	47	0	0	181	252
10	54	92	0	25	324	495
11	107	213	1	0	702	1,023
12	202	399	2	41	1,366	2,010
All	416	790	3	83	2,723	4,015

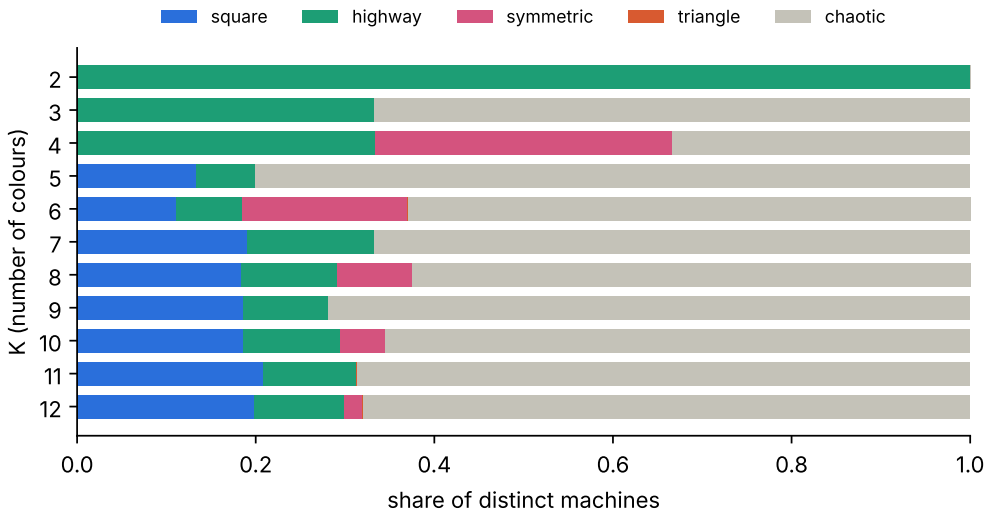


Figure 1: Share of each class among the distinct machines with K colours.

Observation 10 (First letters)**.** For $6 \le K \le 12$, the first three letters lean strongly towards a class:

Rule starts with	Machines	Square	Highway	Chaotic
`LRR`	998	39%	0%	59%
`LRL`	997	18%	7%	75%
`LLR`	998	16%	20%	62%
`LLL`	997	6%	14%	76%

For $K = 3$ to 6, every highway rule starts with `LL`; the first starting with `LR` is `LRLLRRL` ($K = 7$), which leaves B only at step 314,536. The numbers of highways starting `LL` / `LR` for $K = 3, \ldots, 12$ are 1/0, 2/0, 1/0, 2/0, 8/1, 11/2, 21/3, 43/11, 89/18, 163/39.

Observation 11 (Slow growers)**.** 228 machines that are not highways are still under 40 squares wide after 10^6 steps. For example `LLLRR` is 28 squares wide after 10^6 steps, 36 after 1.6×10^7

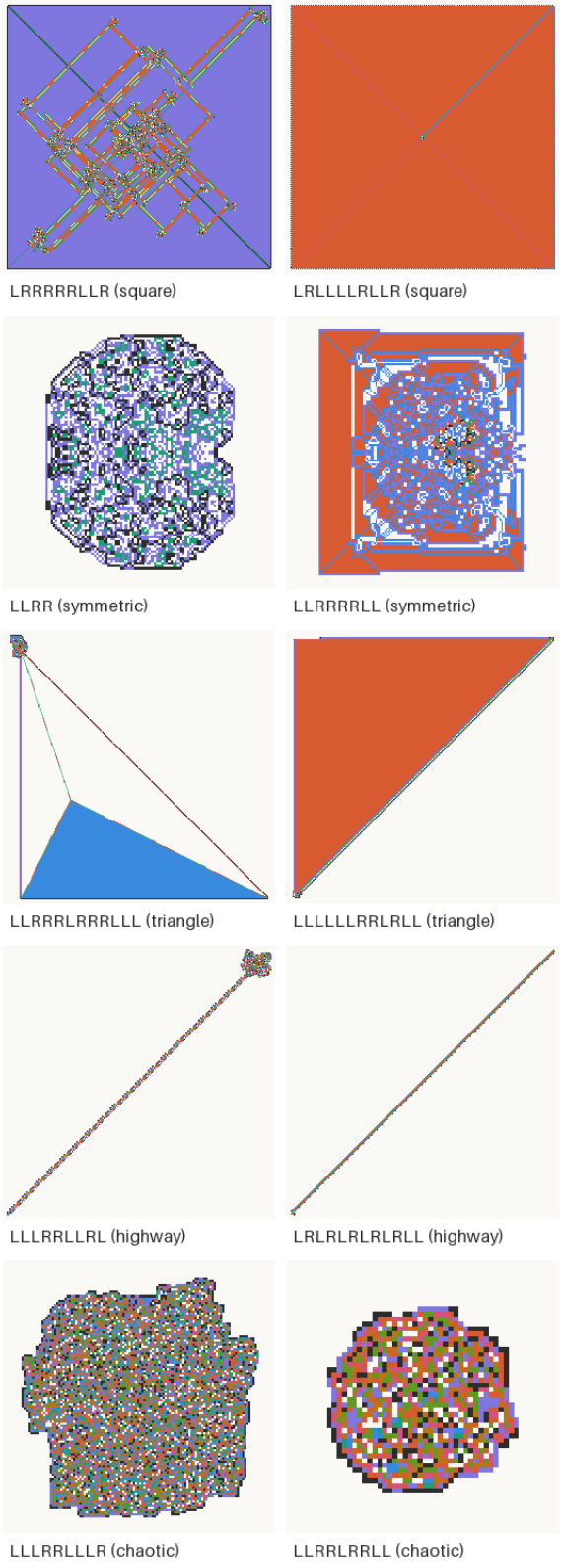


Figure 2: Two machines of each class from the empty grid, cropped to the coloured area (white is colour 0), after 10^6 steps; highways are shown when they leave B.

and 49 after 1.024×10^9, gaining about four squares each time the number of steps is multiplied by four.

5.1 Hidden copies among the highways

The step at which highway machines leave B showed that many of them are the same machine: every highway rule beginning `LLRLLR` leaves B at step 26,969, whatever its length and later letters.

Lemma 12 (Unread letters)**.** *If, up to step T, the ant with rule w visits no square more than $M < |w|$ times, then up to step T it reads only $w_0, \ldots, w_{M-1}$, and every rule with at least M letters that starts with $w_0 \cdots w_{M-1}$ walks the same path up to step T.*

Proof. On its n-th visit to a square the ant reads $w_{(n-1) \bmod |w|} = w_{n-1}$, as $n \le M < |w|$; by induction on the step, every such rule makes the same turns. □

Up to leaving B, 164 of the 416 highway machines never read their last letters, in four families (Table 3); in the first three the letters read are a rule $\mathtt{L}^n\mathtt{R}$ written twice, so the path is that of $\mathtt{L}^n\mathtt{R}$. This is how the family of Theorem 3 first appeared. Grouping machines that read the same letters, there are 252 different highway paths.

Table 3: Highway machines that never read their last letters before leaving B.

Letters read	Same path as	Most visits	Leaves B at	Copies
`LLRLLR`	`LLR` ($K = 3$)	6	26,969	124
`LLLRLLLR`	`LLLR` ($K = 4$)	8	38,949	29
`LLLLRLLLLR`	`LLLLR` ($K = 5$)	10	50,929	6
`LLLRRRLLLR`	itself ($K = 10$)	10	56,928	5

6 Discussion

The proof method. For $\mathtt{L}^{K-1}\mathtt{R}$ the number of colours changes only how many laps the ant spends in each spin, never where it turns right, so one symbolic run covers infinitely many ants. Gajardo, Lutfalla and Rao [9] used the same spins (their elementary cycles) in proofs written by hand; the new points here are that the whole computation is done mechanically with K as a symbol, that it starts from the empty grid, and that its output is a certificate checked by an independent program. The method should apply to other families in which a parameter changes only the length of spins.

Limits of the census. The classes describe each picture after 10^6 steps from the empty grid. A machine called chaotic may build a highway later: the slowest highway found took 937,513 steps to leave B, and [10] shows that some ants mix highways with other behaviours. Two machines checked for 2×10^9 steps, `LLLRL` (a blob 1,954 squares wide after 1.024×10^9 steps) and `LLLRR` (a disc 49 squares wide), built no highway. The tests are rough: a few blobs sit inside square frames, and `LLRLRRLLL` grows a chequered arrowhead that has no class of its own.

Further work. (1) The rare period-156 highway of `LLLR` [11]: in our own search, 20 of 20,000 random 11×11 starts reached it. Its road spans 9 diagonals with 4.5 diagonals' worth of colour, against 5 and 3 for the period-52 road, yet each period takes three times as long; the method here may explain it. (2) Running the chaotic machines much longer. (3) Proving the even-runs pattern. (4) Submitting the count sequences by K to the OEIS once stable.

Code and data availability

All code and data, including the certificate and its checker, are in the repository

`https://github.com/VishnuVaidyanathan/langtons-ant-census`

The script `run_all.sh` rebuilds every number, table and figure in this paper in about two minutes on a laptop. Appendix A lists the files.

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A Repository contents

File	Purpose
`run_all.sh`	Rebuilds everything from scratch.
`proof/symbolic.py`	The symbolic run of $\mathtt{L}^{K-1}\mathtt{R}$ to the edge of B (Proposition 6).
`proof/highway.py`	Checks Theorem 3: conditions (a)–(c), shortest period, exact start.
`proof/write_certificate.py`	Writes the certificate `proof/certificate.txt`.
`proof/check_certificate.py`	Independent checker of the certificate (Section 4.6).
`proof/highway_concrete.py`	Plain step-by-step cross-check of the theorem for chosen K.
`proof/skeleton.py`	Observations behind Proposition 6, for several K.
`src/ant_census.c`	Runs all 2^K rule words for one K; prints shape measures.
`src/classify.py`	Removes repeated words and mirror twins; assigns classes.
`src/analyse.py`	All numbers of Section 5.
`src/ant_visits.c`	Step of leaving B, and most visits to one square, for one rule.
`src/ant_snap.c`, `src/make_figures.py`	Figures.
`data/census.csv`	The 4,015 distinct machines with class and measures.
`data/census_raw.csv`	Measures for all 8,188 rule words.
`data/highways.txt`, `data/squares.txt`	Highway and square machines by K.